

# Yash Chourasia

Applied ML Engineer | XAI & Scalable Backend Systems | FastAPI / PyTorch  
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**Location:** London, United Kingdom | **Target Roles:** Software Engineer, Backend Developer, Data Scientist, Applied AI/ML Engineer

## Professional Summary

Applied ML and backend-focused **MSc Advanced Computer Science** student at the **University of Leicester**, with hands-on experience in **Python, FastAPI, PyTorch, full-stack development, BI pipelines, and production-ready AI systems**. Built real-time business dashboards, scalable backend applications, XAI research pipelines, ML-based cyber-attack detection systems, and AI-enabled web platforms. Strong practical background across **explainable AI, satellite image classification, REST APIs, Power BI, Microsoft Fabric, Flask, MySQL, React.js, and Flutter**. Seeking entry-level roles in software engineering, backend development, data science, or applied AI/ML.

## Core Skills

- **Programming:** Python, Java, SQL, HTML, CSS, JavaScript, React.js, Flutter
- **Backend & Web:** FastAPI, Flask, Node.js, REST APIs, Firebase, Full-Stack Development
- **AI/ML & XAI:** PyTorch, scikit-learn, TensorFlow, Grad-CAM, Grad-CAM++, Integrated Gradients, SmoothGrad
- **Data & BI:** Power BI, Microsoft Fabric, MySQL, Jupyter Notebook, Dashboarding, BI Pipelines, Data Visualisation
- **Application Areas:** Remote Sensing, Cyber-Attack Detection, E-Commerce, AI Chatbots, Business Intelligence
- **Tools & Platforms:** Git, GitHub, Android Studio, VS Code, Agile Collaboration
- **Languages:** English Fluent, Hindi Native, Gujarati, Marathi Conversational

## Experience

### Data Scientist

*SoftSensor AI / Delhi, India*

*Jan 2024 – Dec 2024*

- Delivered BI and analytics solutions for a **Dubai-based client**, supporting sales performance tracking, operational reporting, and business decision-making across a one-year engagement.
- Built **5 Power BI dashboards** backed by **Microsoft Fabric pipelines**, giving business teams real-time visibility into client sales performance.
- Replaced fragmented, manually compiled reports with live dashboards that became the primary source of truth for sales decision-making.
- Eliminated repeated manual reporting cycles, reducing analyst reporting overhead and enabling product and operations teams to act on live data instantly.
- Transformed raw client sales and operational datasets into structured BI-ready models, improving reporting consistency, data accuracy, and stakeholder confidence in business metrics.

## Projects

### Colouring Black Boxes – Visualisation of AI Decision Making

Python, PyTorch, XAI | Feb 2026 – May 2026

*Postgraduate Research, University of Leicester*

- Investigated explainability methods for satellite image classification across **CNNs, Vision Transformers, and Vision Mamba**.
- Implemented **Grad-CAM, Grad-CAM++, Integrated Gradients, and SmoothGrad** using Python and PyTorch.
- Quantified faithfulness, stability, and sensitivity metrics with failure-case analysis to compare XAI suitability across model architectures.
- Found Vision Transformers achieved the highest satellite imagery accuracy, while Vision Mamba was CPU-constrained but showed strong potential for GPU-scaled remote sensing workloads.

### Intelligent Defence System – ML for Cyber Attack Detection

Python, Flask, MySQL, ML | Sep 2023 – Jun 2024

*Cyber-Physical Systems Security Application*

- Built an adaptive ML-based cyber-attack detection system for cyber-physical systems where traditional rule-based security methods were insufficient.
- Architected a model-selection pipeline evaluating **6 machine learning algorithms** sequentially to identify the highest-performing model for real-time threat classification.
- Developed a full-stack **Flask and MySQL** web application with real-time traffic monitoring and scalable attack-pattern classification.
- Achieved **85% detection accuracy** for novel threat patterns with demonstrated scalability to previously unseen attack types.

### E-Commerce Website for Clothing with Integrated AI Chatbot

React.js, Node.js, Firebase | Jun 2022 – Sep 2023

*Full-Stack Retail Platform*

- Designed and built a full-stack clothing e-commerce platform using **React.js and Node.js** with secure payment gateway integration.
- Integrated an AI chatbot to provide real-time customer assistance and improve the end-to-end shopping experience.
- Improved platform security by **25%** and increased overall user satisfaction by **35%**.

## Education

**MSc Advanced Computer Science** — University of Leicester, United Kingdom

*Jan 2025 – July 2026*

**B.E. Computer Engineering, First Class** — St. Francis Institute of Technology, Mumbai University, India

*Jan 2020 – Jun 2024*

**CGPA:** 7.62/10

## Certifications

### HackerRank Certifications

Software Engineer | JavaScript Intermediate | Problem Solving Basic | Python | React | REST API Intermediate | Node.js Basic

*2025*

## Extracurricular Activities

### Rekindle, SFIT Alumni Association, Mumbai

*2023*

- Led **50+ volunteers** for an alumni event with **500+ participants**.

### Community Work

- Led a **200+ sapling plantation drive**, conducted **5 technical workshops** for **100+ students**, and supported women entrepreneurship training for **10 micro-enterprises**.